

Lab-14.2

Course : AI Assisted Coding

Topic:- Web Design Application – AI-Assisted HTML/CSS/JS Generation

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Task-01

Final Optimal Prompt:

Create a simple responsive landing page for a product or service using **HTML and CSS**.

Requirements:

- Include a **header with navigation**, a **hero section with product/service description and call-to-action button**, and a **footer**.
 - Use **modern layout techniques such as Flexbox or CSS Grid**.
 - Suggest proper **layout alignment, spacing, and typography** for better UI design.
 - Ensure the page is **responsive across desktop, tablet, and mobile devices**.
 - Provide clean, well-structured code with comments.
-

Code Screenshot :

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Product Landing Page</title>
7      <link rel="stylesheet" href="styles.css">
8  </head>
9
10 <body>
11
12 <!-- ===== HEADER ===== -->
13 <header>
14     <div class="container nav">
15         <h1 class="logo">SmartProduct</h1>
16
17         <nav>
18             <ul class="nav-links">
19                 <li><a href="#">Home</a></li>
20                 <li><a href="#">Features</a></li>
21                 <li><a href="#">Pricing</a></li>
22                 <li><a href="#">Contact</a></li>
23             </ul>
24         </nav>
25     </div>
26 </header>
27
28 <!-- ===== HERO SECTION ===== -->
29 <section class="hero">
30     <div class="container hero-content">
31
32         <div class="hero-text">
33             <h2>Make Your Work Smarter</h2>
34             <p>
35                 SmartProduct helps teams automate tasks, collaborate faster,
36                 and achieve better productivity with modern tools.
37             </p>
38
39             <a href="#" class="cta-btn">Get Started</a>
40         </div>
41
42         <div class="hero-image">
43             
44         </div>
45     </div>
46 </section>
47
48 <!-- ===== FOOTER ===== -->
49 <footer>
50     <p>© 2026 SmartProduct. All rights reserved.</p>
51 </footer>
52
53
54 </body>
55 </html>

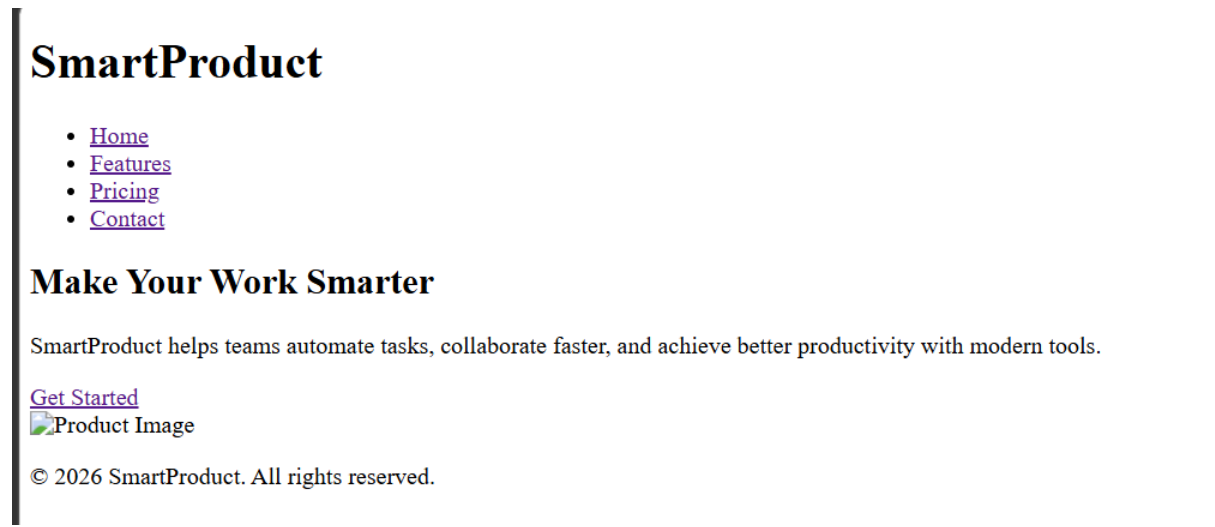
```

```

1  /* ===== GLOBAL STYLES ===== */
2  * {
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7
8  body {
9      font-family: Arial, sans-serif;
10     line-height: 1.6;
11 }
12 /* Container keeps content centered */
13 .container {
14     width: 90%;
15     max-width: 1100px;
16     margin: auto;
17 }
18 /* ===== HEADER ===== */
19 header {
20     background: #111;
21     color: white;
22     padding: 20px 0;
23 }
24 .nav {
25     display: flex;
26     justify-content: space-between;
27     align-items: center;
28 }
29 .logo {
30     font-size: 24px;
31 }
32 .nav-links {
33     list-style: none;
34     display: flex;
35 }
36 .nav-links li {
37     margin-left: 20px;
38 }
39 .nav-links a {
40     color: white;
41     text-decoration: none;
42     font-weight: 500;
43 }
44 /* ===== HERO SECTION ===== */
45 .hero {
46     padding: 80px 0;
47     background: #f5f5f5;
48 }
49 .hero-content {
50     display: flex;
51     align-items: center;
52     justify-content: space-between;
53     gap: 40px;
54 }
55 .hero-text h2 {
56     font-size: 40px;
57     margin-bottom: 20px;
58 }
59 .hero-text p {
60     margin-bottom: 25px;
61     color: #555;
62 }
63 /* Call-to-action button */
64 .cta-btn {
65     background: #0078ff;
66     color: white;
67     padding: 12px 25px;
68     text-decoration: none;
69     border-radius: 5px;
70 }
71 .cta-btn:hover {
72     background: #0056b3;
73 }
74 .hero-image img {
75     max-width: 100%;
76 }
77 /* ===== FOOTER ===== */
78 footer {
79     background: #111;
80     color: white;
81     text-align: center;
82     padding: 20px;
83 }
84 /* ===== RESPONSIVE DESIGN ===== */
85 /* Tablet */
86 @media (max-width: 900px) {
87     .hero-content {
88         flex-direction: column;
89         text-align: center;
90     }
91 }
92 /* Mobile */
93 @media (max-width: 600px) {
94     .nav {
95         flex-direction: column;
96         gap: 10px;
97     }
98     .nav-links {

```

Output Screenshot:



Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

This task creates a simple landing page using HTML and CSS. The page includes three main sections: header, hero section, and footer. The header contains the site title and navigation links, while the hero section highlights the main product message and a call-to-action button. CSS is used to style typography, spacing, and layout for a clean and modern design. Responsive principles such as flexible layout and centered content ensure the page adapts across devices. AI assistance helps improve layout alignment and typography choices, making the interface visually appealing. The final design demonstrates basic UI structure commonly used in real-world web applications

Task-02

Final Optimal Prompt:

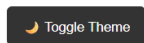
Create a simple web page with a Light and Dark Theme Toggle using HTML, CSS, and JavaScript. The page should include a toggle button, a title, and a short description. Use HTML to structure the layout and CSS to define styles for light mode (white background, dark text) and dark mode (dark background, light text). Implement JavaScript to toggle a class on the body element when the button is clicked, instantly switching between themes. Ensure the design is clean, readable, and uses smooth transitions for better user experience.

Code Screenshot :

index.html charleshtml lab15/index2.html/...

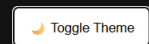
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Light Dark Mode Toggle</title>
7 <style>
8 /* Reset */
9 *{
10 margin:0;
11 padding:0;
12 box-sizing:border-box;
13 }
14 /* Light Theme */
15 body{
16 font-family: Arial, sans-serif;
17 background: # white;
18 color: # black;
19 transition:0.3s;
20 }
21 /* Page Content */
22 .container{
23 width:80%;
24 margin:100px auto;
25 text-align:center;
26 }
27 /* Toggle Button */
28 #theme-toggle{
29 position:absolute;
30 top:20px;
31 left:20px;
32 padding:10px 15px;
33 border:none;
34 border-radius:5px;
35 cursor:pointer;
36 background: # #333;
37 color: # white;
38 }
39 /* Dark Mode */
40 body.dark{
41 background: # #121212;
42 color: # white;
43 }
44 /* Button in dark mode */
45 body.dark #theme-toggle{
46 background: # white;
47 color: # black;
48 }
49 </style>
50 </head>
51 <body>
52 <button id="theme-toggle">🌙 Toggle Theme</button>
53 <div class="container">
54 <h1>Welcome to My Website</h1>
55 <p>
56 This page demonstrates a simple light and dark mode toggle
57 using HTML, CSS, and JavaScript.
58 </p>
59 </div>
60 <script>
61 /* Select button */
62 const toggle = document.getElementById("theme-toggle");
63 /* Toggle theme when clicked */
64 toggle.onclick = function(){
65 document.body.classList.toggle("dark");
66 }
67 </script>
68
```

Output Screenshot:



Welcome to My Website

This page demonstrates a simple light and dark mode toggle using HTML, CSS, and JavaScript.



Welcome to My Website

This page demonstrates a simple light and dark mode toggle using HTML, CSS, and JavaScript.

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

This task creates a simple landing page using HTML and CSS. The page includes three main sections: header, hero section, and footer. The header contains the site title and navigation links, while the hero section highlights the main product message and a call-to-action button. CSS is used to style typography, spacing, and layout for a clean and modern design. Responsive principles such as flexible layout and centered content ensure the page adapts across devices. AI assistance helps improve layout alignment and typography choices, making the interface visually appealing. The final design demonstrates basic UI structure commonly used in real-world web applications

Task-03

Final Optimal Prompt:

Create a **user feedback form** using **HTML, CSS, and JavaScript**.

Form fields:

- Username

- Rating (1–5)
 - Comments
-

Requirements:

- Use **JavaScript** to validate required fields.
 - Display **dynamic success or error messages** after form submission.
 - Improve **form usability, layout, and spacing** for a better user experience.
 - Ensure the form is **responsive and visually clean**.
 - Provide clear and structured code with comments.
-

Code Screenshot :

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>User Feedback Form</title>
7 <link rel="stylesheet" href="style3.css">
8 </head>
9 <body>
10 <div class="container">
11 <h2>User Feedback</h2>
12 <form id="feedbackForm">
13 <!-- Username -->
14 <label for="username">Username</label>
15 <input type="text" id="username" placeholder="Enter your name">
16 <!-- Rating -->
17 <label for="rating">Rating</label>
18 <select id="rating">
19 <option value="">Select Rating</option>
20 <option value="1">1 - Poor</option>
21 <option value="2">2 - Fair</option>
22 <option value="3">3 - Good</option>
23 <option value="4">4 - Very Good</option>
24 <option value="5">5 - Excellent</option>
25 </select>
26 <!-- Comments -->
27 <label for="comments">Comments</label>
28 <textarea id="comments" rows="4" placeholder="Write your feedback"></textarea>
29 <button type="submit">Submit Feedback</button>
30 </form>
31 <!-- Message Area -->
32 <p id="message"></p>
33 </div>
34 <script src="script3.js"></script>
35 </body>
36 </html>
```

```

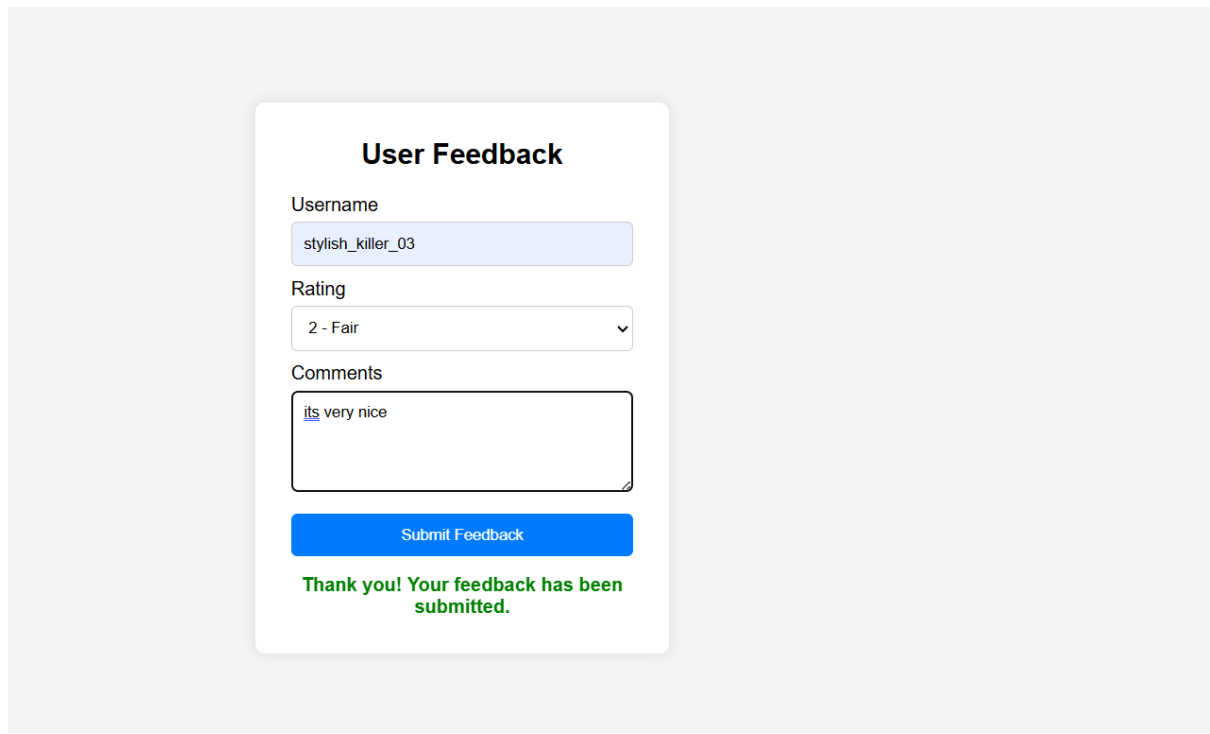
1      *{
2      margin:0;
3      padding:0;
4      box-sizing:border-box;
5      font-family:Arial;
6      }
7
8      body{
9      background: #f4f4f4;
10     display:flex;
11     justify-content:center;
12     align-items:center;
13     height:100vh;
14     }
15
16     /* Form container */
17     .container{
18     background: #fff;
19     padding:30px;
20     width:350px;
21     border-radius:8px;
22     box-shadow:0 0 10px rgba(0,0,0,0.1);
23     }
24
25     h2{
26     text-align:center;
27     margin-bottom:20px;
28     }
29
30     /* Inputs */
31     label{
32     display:block;
33     margin-top:10px;
34     margin-bottom:5px;
35     }
36
37     input, select, textarea{
38     width:100%;
39     padding:10px;
40     border:1px solid #ccc;
41     border-radius:5px;
42     }
43
44     /* Button */
45     button{
46     margin-top:15px;
47     width:100%;
48     padding:10px;
49     background: #007bff;
50     color: #fff;
51     border:none;
52     border-radius:5px;
53     cursor:pointer;
54     }
55
56     button:hover{
57     background: #0056b3;
58     }
59
60     /* Message */
61     #message{
62     margin-top:15px;
63     text-align:center;
64     font-weight:bold;
65     }
66
67     /* Responsive */
68     @media(max-width:500px){
69     .container{
70     width:90%;
71     }
72     }

```



```
1  const form = document.getElementById("feedbackForm");
2  const message = document.getElementById("message");
3
4  Complexity is 5 Everything is cool!
5  form.addEventListener("submit", function(event){
6
7      event.preventDefault();
8
9      const username = document.getElementById("username").value.trim();
10     const rating = document.getElementById("rating").value;
11     const comments = document.getElementById("comments").value.trim();
12
13     /* Validation */
14     if(username === "" || rating === "" || comments === ""){
15
16         message.style.color = "red";
17         message.textContent = "Please fill in all fields.";
18
19         return;
20     }
21
22     /* Success Message */
23
24     message.style.color = "green";
25     message.textContent = "Thank you! Your feedback has been submitted.";
26
27     form.reset();
28
29
30 });
```

Output Screenshot:

A screenshot of a web form titled "User Feedback". The form is white with a light gray border and is centered on a light gray background. It contains three input fields: a text field for "Username" with the value "stylish_killer_03", a dropdown menu for "Rating" with the value "2 - Fair", and a text area for "Comments" with the text "its very nice". Below the input fields is a blue button labeled "Submit Feedback". Below the button is a green message that says "Thank you! Your feedback has been submitted.".

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

This task builds a simple feedback form for collecting user input. The form includes fields for username, rating, and comments. JavaScript validation checks whether required fields are filled before submission. If the fields are empty, an error message appears dynamically without refreshing the page. When valid input is provided, a success message confirms that the feedback was submitted. AI suggestions help improve layout, usability, and input structure for better user interaction. This approach enhances user experience by providing instant feedback and preventing incomplete submissions. Such validation techniques are commonly used in real-world web applications to ensure data accuracy.

Task-04

Final Optimal Prompt:

Create a **responsive image gallery** using **HTML, CSS, and JavaScript**.

Requirements:

- Dynamically **load images using JavaScript**.
- Add **Next and Previous navigation buttons** to browse images.

- Optimize the **layout responsiveness for different screen sizes**.
 - Use clean CSS styling to make the gallery visually appealing.
 - Provide structured and well-commented code.
-

Code Screenshot :

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Responsive Image Gallery</title>
7
8  <link rel="stylesheet" href="style4.css">
9  </head>
10 <body>
11 <div class="gallery-container">
12 <h2>Image Gallery</h2>
13 <!-- Image Display -->
14 <div class="image-box">
15 <img id="gallery-image" src="" alt="Gallery Image">
16 </div>
17 <!-- Navigation Buttons -->
18 <div class="buttons">
19 <button id="prevBtn">Previous</button>
20 <button id="nextBtn">Next</button>
21 </div>
22 </div>
23 <script src="script4.js"></script>
24 </body>
25 </html>
```

```

1  /*{
2  margin:0;
3  padding:0;
4  box-sizing:border-box;
5  font-family:Arial, sans-serif;
6  }
7  body{
8  display:flex;
9  justify-content:center;
10 align-items:center;
11 height:100vh;
12 background:■ #f4f4f4;
13 }
14 /* Gallery Container */
15 .gallery-container{
16 text-align:center;
17 background:■ white;
18 padding:20px;
19 border-radius:8px;
20 box-shadow:0 0 10px ■ rgba(0,0,0,0.1);
21 width:80%;
22 max-width:600px;
23 }
24 /* Image Box */
25 .image-box{
26 margin:20px 0;
27 }
28 .image-box img{
29 width:100%;
30 height:auto;
31 border-radius:6px;
32 }
33 /* Buttons */
34 .buttons{
35 display:flex;
36 justify-content:space-between;
37 }
38 button{
39 padding:10px 20px;
40 border:none;
41 background:■ #007BFF;
42 color:■ white;
43 border-radius:5px;
44 cursor:pointer;
45 }
46 button:hover{
47 background:■ #0056b3;
48 }
49 /* Responsive Design */
50 @media(max-width:600px){
51 .gallery-container{
52 width:95%;
53 }
54 button{
55 padding:8px 15px;
56 }
57 }

```

```
1 // Array of image URLs
2 const images = [
3   "https://picsum.photos/id/1015/600/400",
4   "https://picsum.photos/id/1016/600/400",
5   "https://picsum.photos/id/1025/600/400",
6   "https://picsum.photos/id/1035/600/400",
7   "https://picsum.photos/id/1043/600/400"
8 ];
9 // Get elements
10 const imageElement = document.getElementById("gallery-image");
11 const nextBtn = document.getElementById("nextBtn");
12 const prevBtn = document.getElementById("prevBtn");
13 // Current image index
14 let currentIndex = 0;
15 // Load first image
16 imageElement.src = images[currentIndex];
17 // Next button
18 nextBtn.addEventListener("click", function(){
19   currentIndex++;
20   if(currentIndex >= images.length){
21     currentIndex = 0;
22   }
23   imageElement.src = images[currentIndex];
24 });
25 // Previous button
26 prevBtn.addEventListener("click", function(){
27   currentIndex--;
28   if(currentIndex < 0){
29     currentIndex = images.length - 1;
30   }
31   imageElement.src = images[currentIndex];
32 });
```

Output Screenshot:

Image Gallery

[Previous](#)[Next](#)

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

This task creates a dynamic image gallery using JavaScript. Images are stored in an array, and a variable tracks the currently displayed image index. The next() and prev() functions update the index and load the appropriate image. Navigation buttons allow users to browse through the gallery easily. AI suggestions help optimize the layout for responsive viewing across different screen sizes. The modulo operation ensures smooth looping through images when reaching the first or last item. This approach demonstrates how JavaScript can dynamically update webpage content without refreshing the page, creating a more interactive and engaging user interface.

Tack-05

Final Optimal Prompt:

Create a notification banner at the top of a webpage using HTML, CSS, and JavaScript

Requirements:

Use JavaScript to show and hide the banner.

Style the banner using CSS for a clean and modern look.

Add smooth transitions or animations when the banner appears or disappears.

Ensure the UI behavior is smooth and user-friendly.

Provide clean, well-organized, and commented code.

Code Screenshot :

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Notification Banner</title>
7
8 <link rel="stylesheet" href="style5.css">
9
10 </head>
11 <body>
12
13 <!-- Notification Banner -->
14 <div id="notification" class="banner">
15   <span>🔥 New Offer! Get 30% off on our services.</span>
16   <button id="closeBtn">&times;</button>
17 </div>
18
19 <!-- Button to show banner again -->
20 <div class="content">
21   <h1>Welcome to Our Website</h1>
22   <p>This page demonstrates a notification banner.</p>
23   <button id="showBtn">Show Notification</button>
24 </div>
25
26 <script src="script5.js"></script>
27
28 </body>
29 </html>
```

```

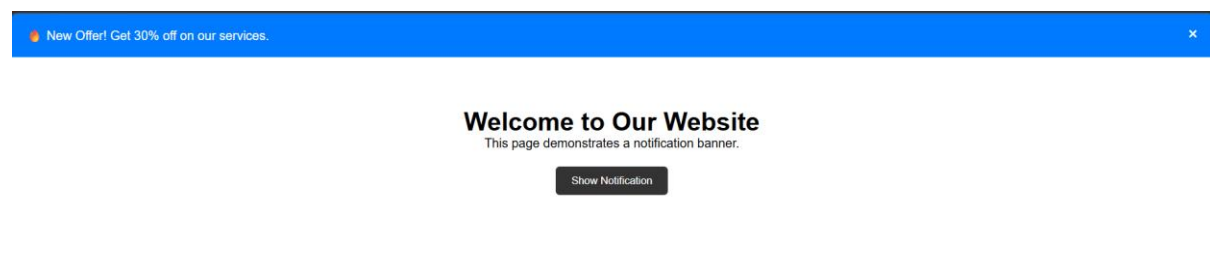
1  /* Reset */
2  *{
3  margin:0;
4  padding:0;
5  box-sizing:border-box;
6  font-family:Arial, sans-serif;
7  }
8  /* Notification Banner */
9  .banner{
10 position:fixed;
11 top:-80px; /* hidden initially */
12 left:0;
13 width:100%;
14 background: #007BFF;
15 color: white;
16 padding:15px 20px;
17 display:flex;
18 justify-content:space-between;
19 align-items:center;
20 transition:top 0.4s ease;
21 z-index:1000;
22 }
23 /* Visible banner */
24 .banner.show{
25 top:0;
26 }
27 /* Close button */
28 .banner button{
29 background:none;
30 border:none;
31 color: white;
32 font-size:22px;
33 cursor:pointer;
34 }
35 /* Page Content */
36 .content{
37 text-align:center;
38 margin-top:120px;
39 }
40 .content button{
41 padding:10px 20px;
42 margin-top:20px;
43 border:none;
44 background: #333;
45 color: white;
46 border-radius:5px;
47 cursor:pointer;
48 }
49 /* Responsive */
50 @media(max-width:600px){
51 .banner{
52 font-size:14px;
53 padding:12px;
54 }
55 }

```



```
1 // Select elements
2 const banner = document.getElementById("notification");
3 const closeBtn = document.getElementById("closeBtn");
4 const showBtn = document.getElementById("showBtn");
5
6 // Show banner when page loads
7 window.onload = function(){
8   banner.classList.add("show");
9 };
10
11 // Hide banner when close button clicked
12 closeBtn.addEventListener("click", function(){
13   banner.classList.remove("show");
14 });
15
16 // Show banner again
17 showBtn.addEventListener("click", function(){
18   banner.classList.add("show");
19 });
```

Output Screenshot:



Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

This task demonstrates how to implement a notification banner on a webpage. The banner appears at the top of the page and can be shown or hidden using JavaScript functions. CSS is used to style the banner and control its initial visibility. When the user clicks the “Show” button, the banner becomes visible, and clicking “Hide” removes it from view. AI assistance helps ensure proper layout positioning and smooth UI behavior. Notification banners are widely used in modern websites to display alerts, announcements, or promotions. This implementation provides a simple and effective way to manage dynamic content visibility.